



VPC ENDPOINTS

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VPC Endpoints

vpce-0d66e6603ac248c86 / NextWork VPC Endpoint

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[Policy](#)

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Details

Endpoint ID

 vpce-0d66e6603ac248c86

VPC ID

 vpc-0fe26ab6abab7822e (NextWork-vpc)

DNS record IP type

service-defined

Private DNS specified domains

–

Status

 Available

Status message

–

IP address type

ipv4

Creation time

Wednesday, 9 September 2026 at 14:21:56 BST

Service name

 com.amazonaws.eu-west-2.s3

Service region

eu-west-2

Endpoint type

Gateway

Private DNS names enabled

No

Private DNS preference

–

Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a networking service provided by AWS to give us the ability to isolate our resources from the public internet, set up connections between our resources and control traffic flow/security.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to setup a VPC endpoint, specifically a S3 Gateway. This provides my VPC with direct, private access to another AWS service.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was that endpoint gateways are only made for Amazon S3 and Amazon Dynamo DB and that other endpoints must use an interface.

This project took me...

This project took me one hour and twenty minutes including documentatio.

In the first part of my project...

Step 1 - Architecture set up

In this step, I will set up the foundations of this project - i.e. launching a VPC, EC2 instance and a S3 bucket. So that I can set up an endpoint architecture and test that setup in the last step of the project.

Step 2 - Connect to EC2 instance

In this step, I am directly connecting to my EC2 instance using EC2 Instance Connect. Connecting to my EC2 instance will help me in accessing S3 and running commands later in this project.

Step 3 - Set up access keys

In this step, I will set up an access key so that my EC2 instance will have access to my AWS environment. you can think of access keys almost like "login details" for EC2 instances/applications (non humans) to interact with your AWS services.

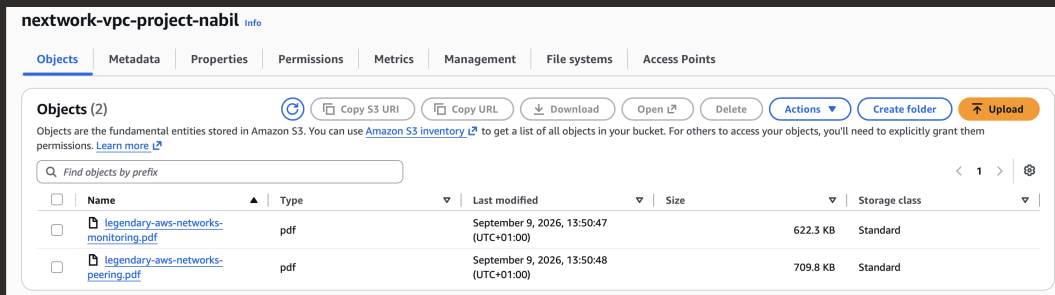
Step 4 - Interact with S3 bucket

In this step, I am applying my access key credentials to my EC2 instance, and then using the AWS CLI and my EC2 instance to access Amazon S3.

Architecture set up

I started my project by launching a VPC with one availability zone and public subnet. I also launched an EC2 instance as well.

I also set up an S3 bucket with two files uploaded inside of it so it makes testing easier.



Access keys

Credentials

To set up my EC2 instance to interact with my AWS environment, I configured the AWS access key ID, secret access key matching that key id, default region type and then the default output format.

Access keys are credentials that an EC2 instance/other servers/applications would need in order to get access to your AWS environment e.g. creating resources, reading what is inside your AWS account etc.

Secret access keys are like passwords in the context of access keys/credential for my EC2 instance to get access to my AWS services/environment.

Best practice

Although I'm using access keys in this project, a best practice alternative is to use IAM admin roles instead. This means any necessary permissions will be attached to an IAM role, and then the role will be associated with the relevant resources.

Connecting to my S3 bucket

The command I ran was 'aws s3 ls'. This command is used to list all s3 buckets in my AWS account that the EC2 instance has access to.

The terminal responded with a list of the S3 buckets on my account. This indicated that the access keys I set up were successful in allowing my EC2 instance to connect to my AWS environment.

```
[ec2-user@ip-10-0-13-24 ~]$ aws s3 ls
2026-09-09 12:50:06 nextwork_vpc-project-nabil
```

Connecting to my S3 bucket

I also tested the command 'aws s3 ls s3://nextwork-vpc-project-nabil' which returned all the objects inside my S3 bucket.

```
[ec2-user@ip-10-0-13-24 ~]$ aws s3 ls s3://nextwork-vpc-project-nabil
2026-09-09 12:50:47      637237 legendary-aws-networks-monitoring.pdf
2026-09-09 12:50:48      726884 legendary-aws-networks-peering.pdf
```

Uploading objects to S3

To upload a new file to my bucket, I first ran the command 'sudo touch /tmp/nextwork.txt'. This command creates an file called nextwork.txt inside the EC2 instance's temp directory.

The second command I ran was 'aws s3 cp /tmp/nextwork.txt s3://nextwork-vpc-project-nabil'. This command will copy this file from the temp directory and move it to the S3 bucket and store it as an object.

The third command I ran was 'aws s3 ls s3://nextwork-vpc-project-nabil' which validated that the file was uploaded into the S3 bucket.

```
[ec2-user@ip-10-0-13-24 ~]$ sudo touch /tmp/nextwork.txt
[ec2-user@ip-10-0-13-24 ~]$ aws s3 cp /tmp/nextwork.txt s3://nextwork-vpc-project-nabil
upload: ../../tmp/nextwork.txt to s3://nextwork-vpc-project-nabil/nextwork.txt
[ec2-user@ip-10-0-13-24 ~]$ aws s3 ls s3://nextwork-vpc-project-nabil
2026-09-09 12:50:47      637237 legendary-aws-networks-monitoring.pdf
2026-09-09 12:50:48      726884 legendary-aws-networks-peering.pdf
2026-09-09 13:14:31           0 nextwork.txt
```

In the second part of my project...

Step 5 - Set up a Gateway

In this step, I will set up a VPC endpoint so that communication between my VPC and other services (especially S3) is direct and secure.

Step 6 - Bucket policies

In this step, I am testing my endpoint connection by blocking off all traffic to my S3 bucket, except from traffic coming from my endpoint.

Step 7 - Update route tables

In this step, I will test my endpoint connection between my bucket and EC2 instance.

Step 8 - Validate endpoint connection

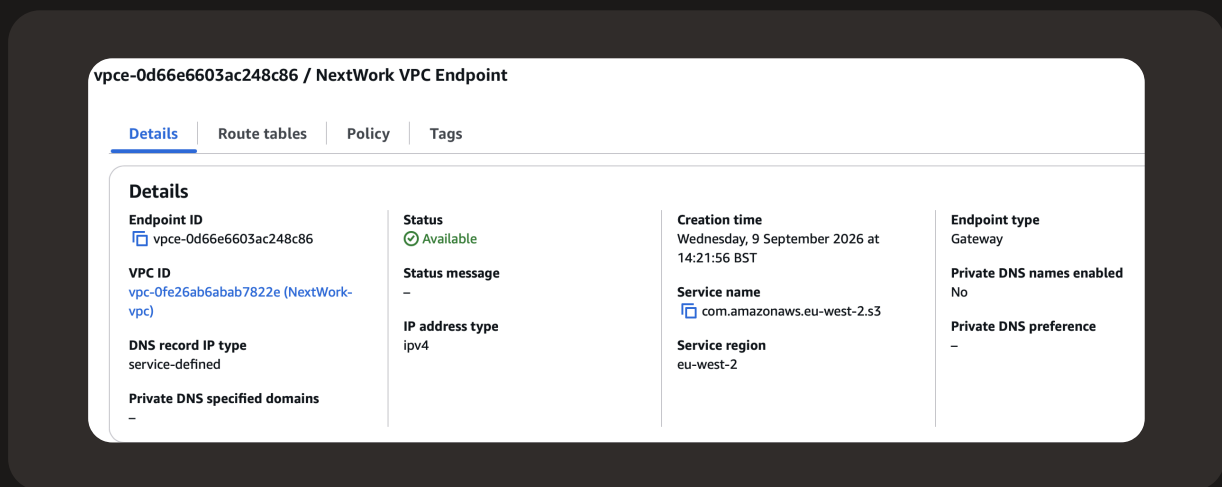
In this step, I will validate my VPC endpoint set up one more time. I am also going to use endpoint policies to restrict my EC2 instance's access to my AWS environment.

Setting up a Gateway

I set up an S3 Gateway, which is a type of endpoint specifically designed for Amazon S3. Gateways work by updating the route table of associated subnets, so that S3 bound traffic goes through the Gateway instead of the public internet.

What are endpoints?

An endpoint is a VPC component that allows your VPC to have a direct connection to AWS services, so that traffic does not have to go through the public internet.



Bucket policies

A bucket policy is a type of policy that has granular control over who has access to an S3 bucket, and what actions they can perform.

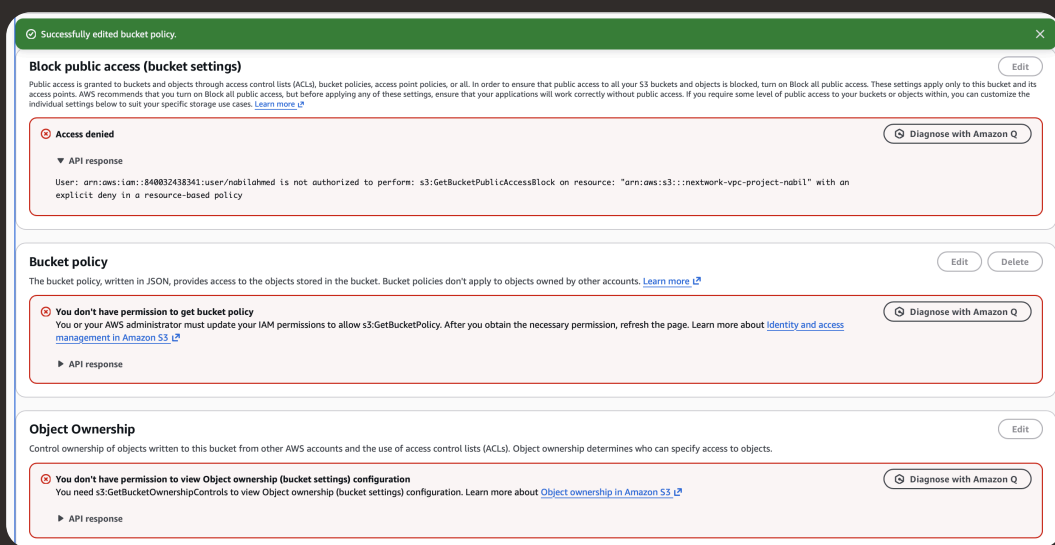
My bucket policy will block all traffic that is not coming from my VPC endpoint from accessing the S3 bucket.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Deny",
      "Principal": "*",
      "Action": "s3:*",
      "Resource": [
        "arn:aws:s3:::nextwork-vpc-project-nabil",
        "arn:aws:s3:::nextwork-vpc-project-nabil/*"
      ],
      "Condition": {
        "StringNotEquals": {
          "aws:sourceVpce": "vpce-0d66e6603ac248c86"
        }
      }
    }
  ]
}
```

Bucket policies

Right after saving my bucket policy, my S3 bucket page showed 'denied access' warnings. This was because my AWS Management Console is trying to access the S3 bucket via public internet but the policy applied to the bucket states that only traffic coming from my VPC endpoint is allowed access.

I also had to update my route table because my route table by default did not provide a route for in my public subnet to the VPC endpoint.



Route table updates

To update my route table, I visited the Endpoints page of my VPC console and I modified the route table from there to associate with my VPC's public subnet.

After updating my public subnet's route table, my terminal could return the list of objects inside my S3 bucket.

Details	Flow logs	Route table	Network ACL	CIDR reservations	Sharing	Tags
Route table: rtb-07c1bb57179acd824 / NextWork-rtb-public						
Routes (3)						
Filter routes						
Destination		Target				
10.0.0.0/16		local				
0.0.0.0/0		igw-0911d8b4c08500bb1				
pl-7ca54015		vpce-0d66e6603ac248c86				

Endpoint policies

An endpoint policy is a type of policy designed for specifying the range of resources and actions permitted by an endpoint.

I updated my endpoint's policy by changing the Effect from "Allow" to "Deny". I could see the effect of this right away, because when I tried running the command to list all the objects stored in my S3 bucket I got a access denied message.

